ZentroMart VoiceOps

Document-grounded customer-support agent with a Gnani voice path

Capstone report | Prepared 23 October 2026 | Based on the official VoiceOps 2026 brief

1. Executive summary

This project packages a document-grounded customer-support agent for the synthetic ZentroMart store. It combines multilingual hybrid retrieval, a local instruct model, structured catalog facts, identity-gated mock order lookup, ticket handover, and an optional Gnani-only speech path. A batch command accepts customer audio files and writes the ASR transcript, agent answer, generated WAV, and retrieved evidence chunks.

44.2%	9.6%	70.2%	55.4%
RAG strict phrase-check pass	Prompt-only baseline	Gold-evidence hit@5	Evidence MRR@5

Visible practice set, 52 questions, Qwen2.5-0.5B-Instruct. Development-set results only.

2. What was built

Step	Brief requirement	Implementation
1. Prompt-only	Use an LLM as a support assistant without company documents.	Same Qwen instruct model and output checks as the RAG run, with no documents. Scored on the 52 visible practice questions: 9.6% strict pass.
2. RAG	Retrieve document chunks, answer from evidence, abstain when unsupported.	Multilingual E5 + BM25 hybrid search with source IDs, authority ranking, catalog lookup, and citation checks.
3. Conversation	Handle follow-ups, clarification, verification, refusal, and handover.	Conversation-local order state; tool-mediated order lookup; one verification retry; ticket success checked before claiming handoff.
4. Voice	Use Gnani STT and TTS only; process folder input and save speech output.	Gnani Realtime SDK path and a batch runner are implemented. A live key/audio run remains unverified in this environment.

> **Note.** The linked W3D5 notebook is an adapter-evaluation lesson for fictional club-event extraction. Its shared helpers demonstrate RAG, but it is not the ZentroMart agent and its event adapter is not used here.

3. Practice evaluation

Step 1 and Step 2 use the same local model and the same 52 visible practice questions. Step 1 receives only a support-assistant prompt. Step 2 receives retrieved evidence and the product record when a specific catalog match is found. The script saves every generated answer and its evidence trace in `evaluation/step_comparison.json`.

Metric	Step 1: prompt-only	Step 2: RAG
Strict phrase-check pass rate	9.6%	**44.2%**
Required-fact phrase pass rate	9.6%	**44.2%**
Forbidden-phrase pass rate	100.0%	**100.0%**
RAG gold-evidence hit@5	Not applicable	**70.2%**
RAG evidence MRR@5	Not applicable	**55.4%**

Evaluation model: Qwen/Qwen2.5-0.5B-Instruct. Questions scored: 52. Evidence-eligible questions: 47.

On these visible practice cases, the Step 2 strict heuristic pass rate was 34.6 percentage points higher than Step 1. This is a development-set comparison, not hidden-test accuracy.

Interpretation and limitations

Required and forbidden facts are scored by literal phrase matching against the supplied annotations. This can mark a correct paraphrase wrong or accept a sentence that mentions a phrase without being semantically correct. Gold-evidence hit@5 measures whether annotated sources appear in the retrieved list; MRR measures the first gold source rank. Review generated answers and misses manually before drawing quality conclusions.

The provided comparison script fixes the model, prompt, question list, and top-k so Step 1 and Step 2 differ by document retrieval. No organizer-held-out questions or audio are included in this report.

Failures found and fixes made

The first local run exposed answers with no usable source citation, so the agent could not safely present them as grounded. The answer path was changed to validate citations and, when the model omits one, quote a short retrieved passage with its real section ID. Query expansion was also added for delivery destinations and damage reports. The final phrase-check score remains 44.2%, and retrieval misses remain; next, use a stronger model or reranker, inspect each failed question, and rerun the complete practice set.

What the comparison shows

Retrieval is the main driver of the gain: adding documents lifts the strict pass rate from 9.6% to 44.2%. The gap between the 70.2% gold-evidence hit@5 and the 44.2% pass rate suggests that many answers are lost after the right evidence is found, which points to the small 0.5B model as a bottleneck as much as retrieval. The two biggest planned improvements are therefore a stronger generator and a reranker.

4. Conversation, tools, and voice

Retrieval and answer path

The parser reads section IDs and front matter from the supplied Markdown KB, skips support logs, and retains the source passage text. The retriever fuses BM25 keyword ranking with multilingual E5 similarity, boosts higher-authority policy and agent-guide material, and collapses overlapping windows by source section. The answer prompt requires brief language-matched replies, source citations, and abstention when evidence is inadequate. Product catalog facts are loaded from CSV rather than generated by the LLM.

Order and escalation path

For an order question, the mock tool is first called without proof and returns no order facts. The assistant asks for the registered phone suffix or delivery pincode, permits one retry, and adds only an allowlisted order summary after verification. Session state supports follow-up questions. The assistant creates a ticket before saying handover succeeded; the ticket summary uses an issue category instead of copying free-form customer text. Explicit human, legal, safety, repeat-contact, high-value damage, and overdue-refund cases have escalation rules.

Policy date nuance

Current documents rank above older material. The two preserved version-3 electronics return clauses are eligible only for a verified electronic item delivered before 1 August 2026. All calculations use the dataset reference date of 20 October 2026, business days exclude Sundays and listed holidays, and return windows are calendar days.

Gnani voice path

The input audio is converted locally to mono 16 kHz PCM, then passed to Gnani Realtime STT. The transcript goes to the same conversation agent. The text reply is normalized for spoken dates, currency, order numbers, and citations before Gnani Realtime TTS creates a WAV. The batch command processes supported audio files recursively, preserves state for files named with a shared `_turn_01`, `_turn_02` stem, and writes transcript, answer, audio, tool/decision trace, retrieved IDs, and retrieved chunk text.

> The source implementation follows Gnani's official STT Realtime and TTS Realtime SDK documentation. The configured TTS model is `timbre-v2.5` and the default voice is Kaveri. No alternative speech provider is included.

5. Reproduction and submission status

Reproduction

Install Python 3.10–3.12 dependencies with `python -m pip install -r requirements.txt`. Extract the official capstone dataset outside the code repository; it must contain `kb/`, `data/`, `practice/`, and `tools/`. Copy `.env.example` to a local `.env` and add a Gnani API key as `GNANI_API_KEY`. The batch runner accepts an audio folder and saves text, speech, and source chunks:

```
python run.py --audio-dir input_audio --output-dir sample_outputs --data-dir
"C:/path/to/Capstone_Project_Dataset_FTTT_2026" --stt-language en-IN --tts-language auto
```

The submission repository is arranged as `README.md`, `.env.example`, `requirements.txt`, `run.py`, `src/`, `report.pdf`, `demo.mp4`, and `sample_outputs/`. The ZIP excludes the dataset copy, virtual environment, model cache, and secrets.

Current status

Status	Item
Passed	Dataset structure validator passes the supplied 9 CSV tables, 21 KB documents, keys, joins, order math, evidence tags, and verification checks.
Passed	Python syntax compilation and batch runner help/argument inspection pass. These checks do not prove live model or voice correctness.
Not verified	Live Gnani STT/TTS, speech quality, and the batch run on a self-recorded sample are not yet verified in this report.
Pending	The two-minute demo video has not been recorded yet. It will use my own voice and only synthetic practice data, showing a grounded answer with a follow-up and a handover or abstention.
Pending	The local Git repository has not been published yet. Its public GitHub URL will be added to `README.md`.

Bonus features

The implementation adds multilingual semantic retrieval, session follow-ups, verified mock order lookup, ticket handover, and spoken formatting for dates and amounts. These go beyond the four required steps.

Known limits

- The supplied order identity check and ticket writer are educational mocks, not production services.
- Phrase checks are heuristic. The local language model can miss retrieval nuance or generate incorrect claims, so review every practice failure.
- The live speech path depends on a valid personal Gnani key, credits, supported audio, and a self-recorded sample.
- No hidden-test score is claimed.

> Dataset, policies, tools, and visible practice examples are synthetic course materials from VoiceOps 2026. No real customer data or recordings are used.